Bankruptcies on supplier-customer network and the 2011 Tohoku Earthquake

Shinya KAWATA[1], Yoshi FUJIWARA[2]

[1,2] Graduate School of Simulation Studies, University of Hyogo, Kobe 650-0047 Japan

E-mail: [1]shinya.no14@gmail.com

Abstract

The 2011 Tohoku Earthquake and Tsunami in the prefectures of Tohoku triggered failures of firms and suspension of corporate activities, which had eventually caused about chained failures. A well-known case is the Toyota and its suppliers in the area, but less known is how one failure caused others for many small and intermediate firms on the supplier-customer network. We analyze and quantify the failures in a year right after the earthquake by using an exhaustive list of bankruptcies and a compilation of investigations. We show that the chains of failures had a tremendous effect in the system.

Keyword: bankruptcies, supplier-customer network, chained failures, The Tohoku Earthquake

Summary

The Tohoku Earthquake and Tsunami in Japan took place in March 2011, which had caused and have been causing failures of firms. It is well-known that a number of suppliers in the Tohoku region were damaged and partly went into bankruptcies, and also that diminishing demand by customers in the area caused bankruptcies of suppliers [1]. According to a list of bankruptcies, one can measure the number of failures in each geographical regions and industrial sectors. Regardless of the metropolitan area, the business relationships of bankrupted firms are wide, but not restricted in neighboring prefectures. In addition, the business relationship in the same prefecture some exception (Kanagawa Prefecture and Chiba Prefecture) has become the most. The list covers more than 20,000 bankruptcies in the one year from April 2011 to March 2012. It would be ideal for understanding mechanisms of failures to examine how a part of the failures were caused by the disaster, and how one failures had caused other failures in the supplier-customer network. By using the data of bankruptcies, the data of their suppliers and customers (about 35,000 links from the failed nodes), and also compiled description of causes in most of the failures [2], we analyze and quantify the failures in the Tohoku region. We show that the chains of failures had a tremendous effect in the system.

References

[1] Ministry of Economy, Trade and Industry, The Small and Medium Enterprise Agency, “White Paper on Small and Medium Enterprises in Japan” (2011)

[2] Data Source: Tokyo Shoko Research, Inc.

Figure

The failures (red nodes of 123 firms) in the Fukushima Networks, one of the most damaged place, and their suppliers or customers (blue nodes of 361 firms are not failure). Companies data in Fukushima prefecture is present 290 companies. Among them, the failures are 61 companies, non-failure are 229 companies. Each arrow from a node A to B represents that A is a supplier or business partner, shareholder of B.

